EXP 5:

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

CODE:

```
clc;
clear;
close all;

% Input Parameters
f0 = 10;
Z0 = 50;
f = 8:0.01:12;

% Directional Coupler S-Parameters
S11 = -25 + 2*((f-f0)/2).^2;
S21 = -(0.6 + 0.1*((f-f0)/2).^2);
S31 = -(10 + 3*((f-f0)/2).^2);
S41 = -(28 - 1.5*((f-f0)/2).^2);

% Coupler Performance
Coupling = -S31;
Isolation = -S41;
IL = -S21;
Directivity = Isolation-Coupling;

% Two-Way Power Divider
S11_div = S11;
S21_div = -(3.0103 + 0.15*((f-f0)/2).^2);
S31_div = S21_div;
S23_div = -30 + 2*((f-f0)/2).^2;

P2 = 10.^(S21_div/10);
P3 = 10.^(S31_div/10);

% Results at 10 GHz
[~,idx] = min(abs(f-f0));

fprintf('\nMICROWAVE COUPLER RESULTS\n');
fprintf('S11 = %.3f dB\n',S11(idx));
```

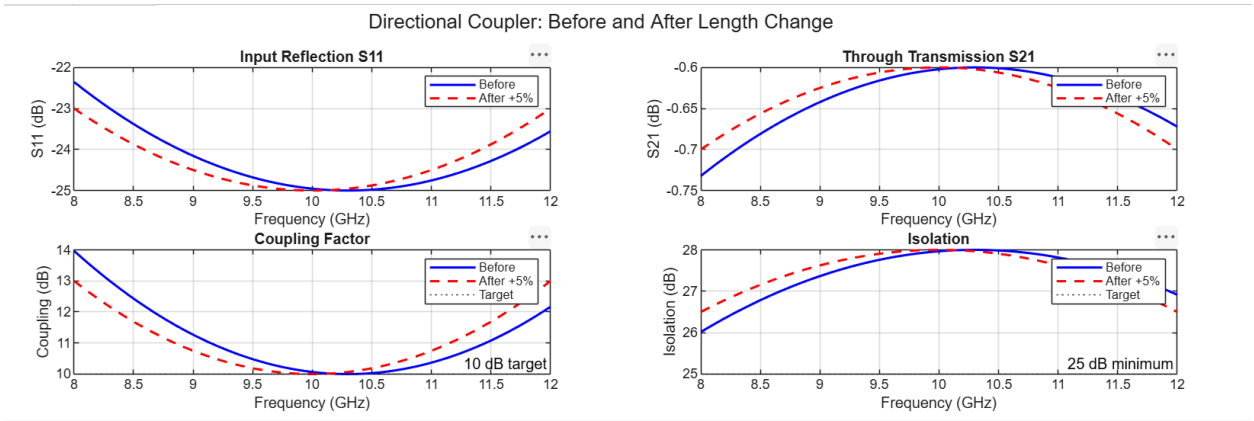
```
fprintf('S21 = %.3f dB\n',S21(idx));
fprintf('S31 = %.3f dB\n',S31(idx));
fprintf('S41 = %.3f dB\n',S41(idx));
fprintf('Coupling = %.3f dB\n',Coupling(idx));
fprintf('Isolation = %.3f dB\n',Isolation(idx));
fprintf('Insertion Loss = %.3f dB\n',IL(idx));
fprintf('Directivity = %.3f dB\n',Directivity(idx));

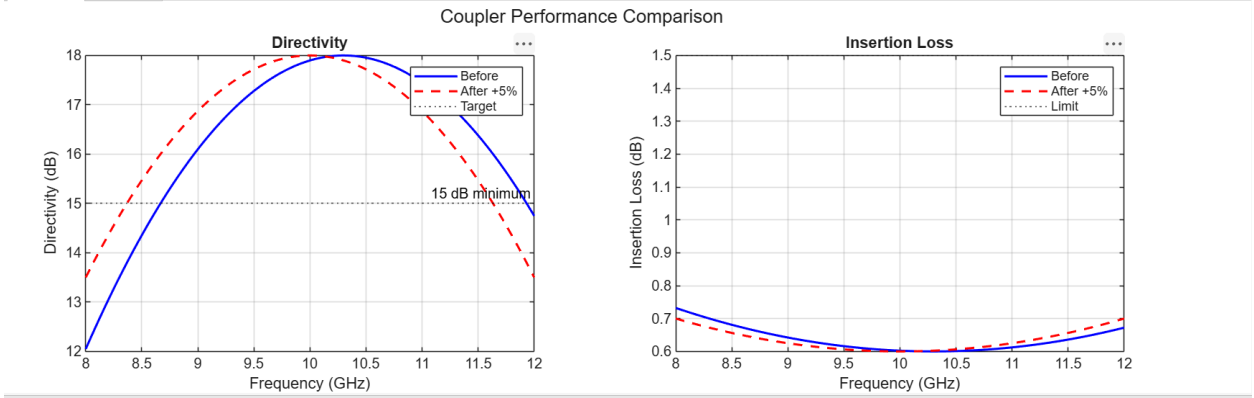
fprintf('\nPOWER DIVIDER RESULTS\n');
fprintf('S11 = %.3f dB\n',S11_div(idx));
fprintf('S21 = %.3f dB\n',S21_div(idx));
fprintf('S31 = %.3f dB\n',S31_div(idx));
fprintf('S23 = %.3f dB\n',S23_div(idx));
fprintf('Output 2 Power = %.3f\n',P2(idx));
fprintf('Output 3 Power = %.3f\n',P3(idx));

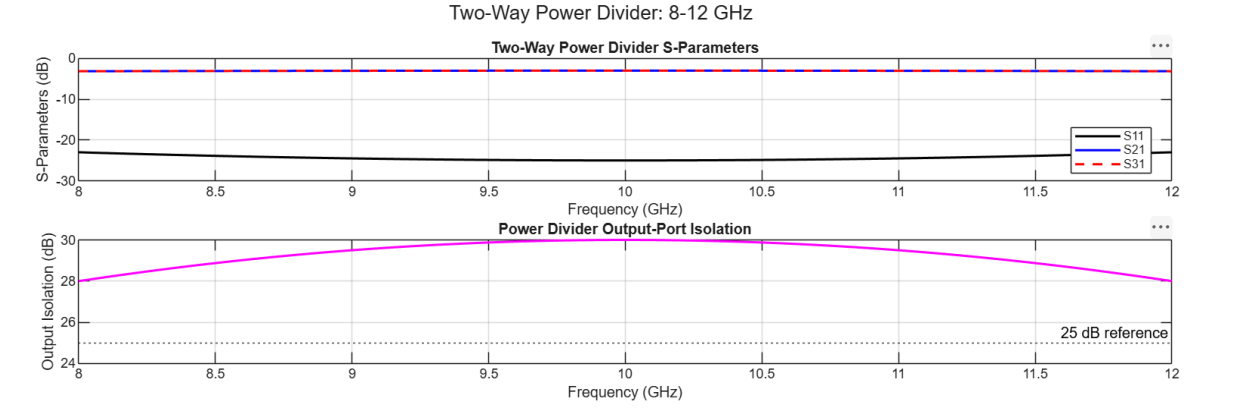
% Plot
figure;
plot(f,Coupling,'b',f,Isolation,'r', ...
    f,IL,'g','LineWidth',1.5);
xlabel('Frequency (GHz)');
ylabel('Performance (dB)');
title('Directional Coupler Performance');
legend('Coupling','Isolation','Insertion Loss');
grid on;

figure;
plot(f,S11_div,'k',f,S21_div,'b', ...
    f,S31_div,'r','LineWidth',1.5);
xlabel('Frequency (GHz)');
ylabel('S-Parameters (dB)');
title('Two-Way Power Divider');
legend('S11','S21','S31');
grid on;
```

OUTPUT:


Directional Coupler: Before and After Length Change
Input Reflection S11
S11 (dB)
Frequency (GHz)
Before
After +5%
Through Transmission S21
S21 (dB)
Coupling Factor
Coupling (dB)
Target
10 dB target
Isolation
Isolation (dB)
25 dB minimum


Coupler Performance Comparison
Directivity
Directivity (dB)
Frequency (GHz)
Before
After +5%
Target
15 dB minimum
Insertion Loss
Insertion Loss (dB)
Limit


Two-Way Power Divider: 8-12 GHz
Two-Way Power Divider S-Parameters
S-Parameters (dB)
Frequency (GHz)
S11
S21
S31
Power Divider Output-Port Isolation
Output Isolation (dB)
25 dB reference

VERIFICATION:

$|S_{31}| = 0.3162$

Isolation $I = -20\log_{10}|S_{41}|$

$25 = -20\log_{10}|S_{41}|$

$$\boxed{|S_{41}| = 0.05623}$$

$IL = -20\log_{10}|S_{21}|$

$0.6 = -20\log_{10}|S_{21}|$

$$\boxed{|S_{21}| = 0.9333}$$

Directivity $D = 1 - C$

$= 28 - 10$

$$\boxed{D = 18\,dB}$$

2 way Power division

$$\frac{P_2}{P_{in}} = \frac{P_3}{P_{in}} = 0.5$$

$P_{dB} = 10\log_{10}(0.5)$

$$\boxed{P_{dB} = -3.0103\,dB}$$